

OFFLINE PREPARATION AND DEPLOYMENT

102-run dataset



run-level train/validation split

Nonlinear AGV model



implemented discrete transition

training/validation truth is not used online

Train · calibrate · linearize



offline identification pipeline

W

Weights + stats

ModernTCN deployment

E

Error moments

P_m, P_o, C by regime

L

LPV database

4125 local models

ONLINE PERCEPTION → CONTROL LOOP

Onboard sensing



22-D feature bank



z_k

$z \Delta_1 \Delta_2 \Delta_4$

ModernTCN-Delta



slope + confidence

θ^M, c^{main}

Qualified observer



IMU · innovation qualification

IMU

θ^{obs}, q_{obs}

Uncertainty-adaptive fusion

NIS gate · bounded ± 0.5

Slope conditioning



deadband · clip ± 10

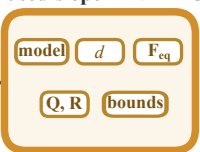
$T_s = 0.01$ s · one update per control period

Reference path
motion command



r^{ref}

Fused-slope LPV-MPC



u_k

AGV plant



y_{k+1}

— temporal primary

— observer auxiliary

— offline deployment